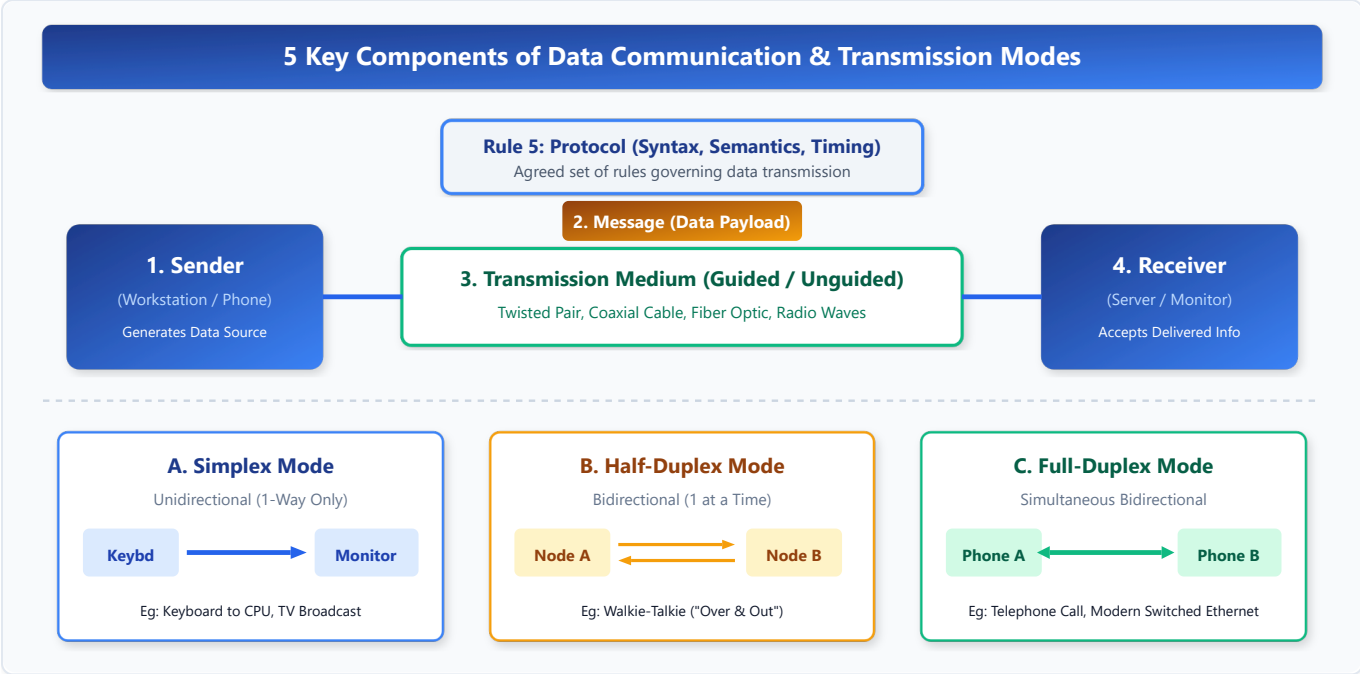


Module 01: Data Communication & Network Fundamentals

AKTU BCS603 / GATE CS Core Foundation: Data communication components, transmission modes, connection types, and network criteria.



1. 5 Core Components of Data Communication

Data communication do ya do se adhik devices ke beech information ka physical transfer hai. Iske 5 essential building blocks hote hain:

- 1. **Message:** Actual data payload (Text, Numbers, Audio, Video, Binary frames).
- 2. **Sender:** Source node jo message transmit karta hai (Workstation, Camera, IoT Sensor).
- 3. **Receiver:** Destination node jo data accept karta hai (Server, Monitor, Actuator).
- 4. **Transmission Medium:** Physical communication path (Guided: Twisted pair, Fiber optic; Unguided: RF, Microwave).
- 5. **Protocol:** Agreed rules and conventions governing syntax, semantics, and synchronization.

2. Transmission Modes Comparison

Parameter	Simplex	Half-Duplex	Full-Duplex
Direction	Strictly Unidirectional (1-way)	Bidirectional, but 1-direction at a time	Simultaneous bidirectional (Both ways at once)
Channel Bandwidth	100% dedicated to single sender	Shared time-wise between participants	Dedicated Tx and Rx paths or frequency split
Turnaround Delay	None	Present (carrier switching / "Over")	Zero turnaround delay
Industry Example	Keyboard to PC, Radio Broadcast	Walkie-Talkie, CB Radio	Telephone Network, Switched Gigabit Ethernet

3. Types of Connections

- **Point-to-Point:** Dedicated physical link connecting exactly two devices. Link bandwidth 100% reserved. (e.g., Leased lines, satellite uplink/downlink).
- **Multipoint (Multidrop):** Single shared communication medium tapped into by multiple stations. Requires MAC arbitration protocols (CSMA/CD, Token passing) to prevent collisions.

AKTU Exam Tip: Jab bhi transmission mode par question aaye, Simplex, Half-Duplex, aur Full-Duplex ka directional diagram aur Walkie-Talkie vs Phone ka example zaroor draw karein!